RATING METHODOLOGY

16 May 2025

TABLE OF CONTENTS

Scope 1
Sector overview 1
Rating approach 3
Trading companies scorecard 5
Discussion of the scorecard factors 8
Other considerations 11
Using the scorecard to arrive at a scorecard-indicated outcome 13
Assigning issuer-level and instrument-level ratings 15
Key rating assumptions 15
Limitations 15
Appendix: Additional financial adjustments for commodity trading companies 16
Moody's related publications 17

Contacts

Karen Nickerson +1.212.553.4924
Associate Managing Director
karen.nickerson@moodys.com

Mariko Semetko +81.3.5408.4209
Associate Managing Director
mariko.semetko@moodys.com

CLIENT SERVICES

Americas 1-212-553-1653
Asia Pacific 852-3551-3077
Japan 81-3-5408-4100
EMEA 44-20-7772-5454

Rating Methodology

Trading Companies

This methodology is no longer in effect. For more information on rating methodologies used by Moody's Investor Services, visit https://ratings.moodys.com/rating-methodologies.

This rating methodology replaces the *Trading Companies* methodology published on 23 June 2022. While this methodology reflects many of the same core principles as the 2022 methodology, we have made changes to the scorecard to provide greater consistency of approach across our corporate sector methodologies. In the Leverage and Coverage factor, we have removed the Debt/Book Capitalization sub-factor and replaced the sub-factors Net Debt/EBITDA and FFO/Debt with Debt/EBITDA and RCF/Net Debt, respectively. We have changed some scoring thresholds. We have also made limited editorial updates to some sections of the methodology, and we have changed the presentation of the scorecard.

OUTDATED METHODOLOGY

Scope

This methodology applies to general and commodity trading companies globally. The companies have a wide range of products and business models.

We define trading companies as companies that are primarily engaged in the trading of commodities or goods, which for Japanese trading companies can include a wide range of products. Commodity trading companies typically have substantial physical inventories of commodities, logistics assets and varying degrees of vertical integration and investments in manufacturing or processing operations.

The determination of a company's primary business is generally based on the preponderance of the company's business risks, which are usually proportionate to the company's revenue, earnings and cash flows.

Sector overview

General trading companies share some characteristics with commodity trading companies but tend to have more diversified global investments at many different points in various global production and supply chains.

General trading companies (GTC)

These essential Japanese trading companies (or as they are referred to in Japan, "sogo shosha") have an important position in the Japanese economy and very close relationships with key domestic banks. The broad credit environment in Japan has tended to be supportive of such prominent companies. These credit considerations are not fully reflected in the characteristics scored in the scorecard, which is designed to apply globally. However, our

ratings incorporate these favorable credit considerations as part of our qualitative assessment of each company.

The business model of general trading companies is highly complex involving a diverse array of earnings in many segments, but these businesses are uniquely shaped to include both traditional trading on the one hand and long-term business investment on the other. Their credit profiles overall are hybrids, lying between that of a corporate and financial institution. As such, a higher level of leverage can be reconciled with a higher rating than would be the case for most purely nonfinancial corporate entities.

At the same time, this group of companies usually carries a large proportion of highly liquid stock and short-term investments, as well as cash and this acts as a strong liquidity buffer that can be used in times of stress to rapidly pay down debt. Due to the diversity of their long-term investments, it is expected that they will also be able to sell – at any time – a portion of their portfolio, or at least a number of intermediate or larger scale investments, should they need to. Financial leverage can be offset to an extent by this financial and operating flexibility.

General trading companies' earnings and cash flow streams are generally divided into three categories: (i) earnings from trading activities, (ii) dividend income from non-controlling interest in investments and (iii) earnings from manufacturing, sales, and financial services mainly conducted by their subsidiaries in which they have controlling interests.

However, those investments are characterized as long-term, less liquid and as having higher risk profiles compared with the trading business. Also, general trading companies have made large investments in the resources and commodities sectors such as iron ore, coal, copper, aluminum, and natural gas, which face cyclical downturns. Therefore, risk management is very important; but due to limited public disclosure, it is challenging to compare and quantify their risk management capabilities. The earnings from their subsidiaries in which they have controlling interests also provide meaningful contributions to their overall profitability. These subsidiaries range from manufacturing to services to leasing, and have different business risk profiles and margins, depending on the type of businesses.

Derivatives are mainly used to hedge market risks related to their exposures such as foreign currency exchange rates, interest rates and commodity prices. Their involvement in proprietary trading is very limited.

Due to the nature of the finance function performed by general trading companies such as leasing and captive finance related businesses, as well as providing short-term financing associated with their trading business, general trading companies are typically highly levered as measured by debt to EBITDA.

Finally, liquidity management is also very important for general trading companies. They generally hold large amounts of cash, which is enough to cover short-term debt obligations. Also, they tend to have committed lines of credit mainly from Japanese banks.

Commodity trading companies (CTC)

Commodity Trading companies are characterized by relatively thin margins, stable demand and volatile credit metrics. In general, these companies make extensive use of derivatives to reduce their exposure to changes in commodity prices and "lock in" their margins. However, they also engage in proprietary trading activities to enhance margins. Specific business and competitive risks vary depending on the commodities sold and the countries in which they operate. The companies can be divided into two segments:

» Larger more diversified companies with vertically integrated operations and investments

» Smaller merchandisers with more limited assets and very limited vertical integration

Logistics assets are important: For traders of most commodities (crude oil and related petroleum or natural gas products, along with electrical energy, can be exceptions), logistics assets are required to support the merchandising of commodities. These assets include silos, warehouses, storage tanks, rail cars, barges, ships and port facilities. These assets can be owned or leased, but typically a company must increase its storage and logistics assets to generate sustainable sales growth.

Use of derivatives: Another unique aspect of this industry is the extensive use of derivatives to limit price risk, as well as the large dollar value of forward purchase and sale contracts these companies enter into. The use of derivatives is so extensive that most companies have direct access to several exchanges around the world.

OUTDATED METHODOLOGY

Proprietary trading: These companies utilize the knowledge they garner from purchasing and selling physical commodities to take long, short or spread positions on specific commodities. The degree of proprietary trading is highly dependent on management's risk tolerance, the compensation structure and the adherence to specific controls (VaR limits, etc.). Many companies have traders that are highly compensated based on their ability to generate profits. Hence, we focus on risk management practices and the controls that are in place at these companies.

Liquidity: Liquidity is of paramount importance for these companies as it can greatly impact their ability to take on additional business, especially when commodity prices are rising or are at elevated levels. Additionally, the market's perception of a company's liquidity can greatly impact its ability to hedge its price exposure on commodities where there is significant basis risk relative to exchange-traded options. Companies in this industry must proactively manage their liquidity to a much greater degree than companies in other industries.

Vertical integration: Most larger companies in the industry are vertically integrated through wholly-owned operations that produce the commodity, or process it into value-added products (which can also be an exchange-traded commodity - soybean oil, soybean meal, flour etc.). We believe that vertical integration provides greater "optionality" and greater stability in margins over time.

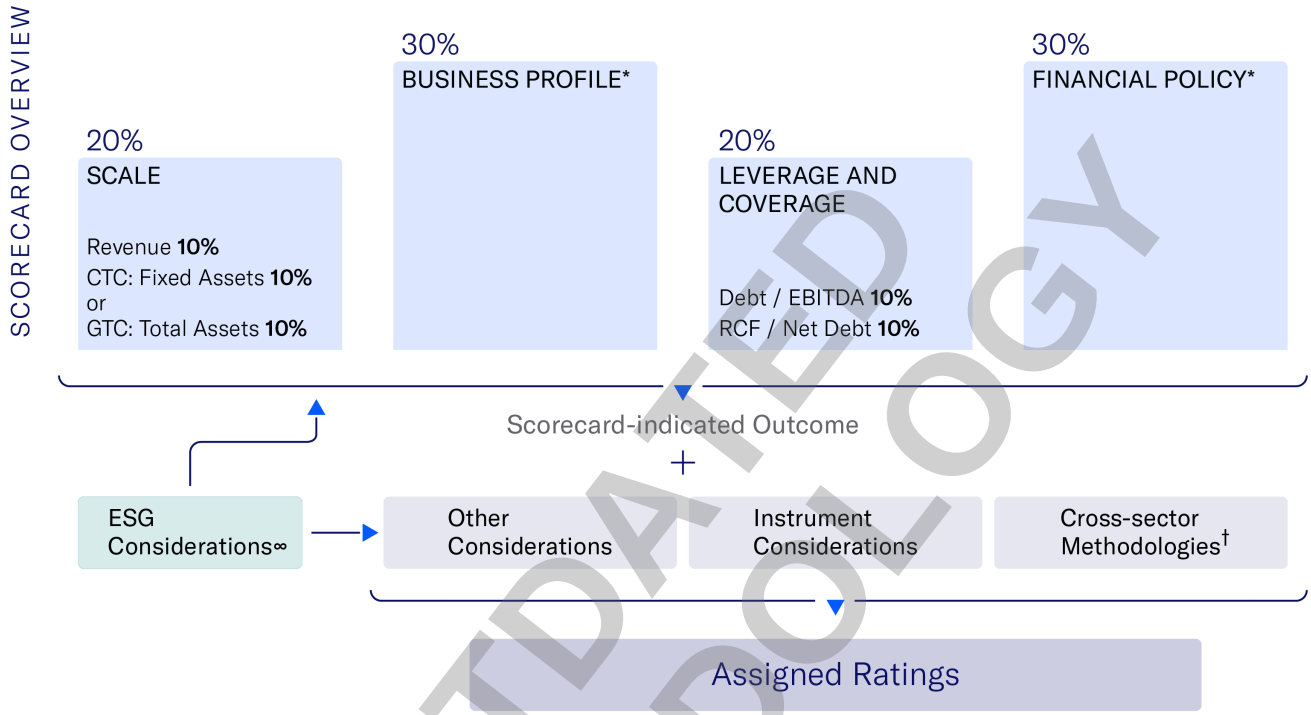
Rating approach

In this rating methodology, we explain our general approach to assessing credit risk of trading companies globally, including the qualitative and quantitative factors that are likely to affect rating outcomes in this sector. We seek to incorporate all material credit considerations in ratings and to take the most forward-looking perspective that visibility into these risks and mitigants permits.

The following schematic illustrates our general framework for the analysis of trading companies, which includes the use of a scorecard. The scorecard-indicated outcome is not expected to match the actual rating for each company. For more information, see the "Other considerations" and "Limitations" sections.

OUTDATED METHODOLOGY

Exhibit 1
Illustration of the trading companies methodology framework



OUTDATED METHODOLOGY

* This factor has no sub-factors.
∞ Environmental, social and governance (ESG) considerations, including, where available, our opinions of exposure to them as expressed in Issuer Profile Scores (IPSs), may affect scorecard factors and other considerations outside of the scorecard. For more information, see the "Other considerations" section.
† Some of the methodological considerations described in one or more cross-sector methodologies may be relevant to ratings in this sector. A link to a list of our sector and cross-sector methodologies can be found in the "Moody's related publications" section.
Source: Moody's Ratings

Trading companies scorecard

For general information about how we use the scorecard and for a discussion of scorecard mechanics, please see the "Using the scorecard to arrive at a scorecard-indicated outcome" section. The scorecard does not include or address every factor that a rating committee may consider in assigning ratings in this sector. Please see the "Other considerations" and "Limitations" sections.

Exhibit 2

Trading companies scorecard

	GTC Weight	CTC Weight	Aaa	Aa	A	Baa	Ba	B	Caa	Ca
Factor: Scale (20%)										
Revenue (USD Billion)[1]	10%	10%	≥ $250	$100 - $250	$50 - $100	$20 - $50	$10 - $20	$1 - $10	$0.5 - $1	< $0.5
Total Assets (USD Billion)[2]	10%	0%	≥ $200	$150 - $200	$100 - $150	$50 - $100	$25 - $50	$10 - $25	$1 - $10	< $1
Fixed Assets (USD Billion)[3]	0%	10%	≥ $75	$30 - $75	$10 - $30	$5 - $10	$1 - $5	$0.25 - $1	$0.1 - $0.25	< $0.1
Factor: Business Profile (30%)										
Business Profile	30%	0%	GTC: Highly diversified global operations. No concentrations in geography, end-market or customer. Expected to have extremely stable and high earnings evidenced by ROA of at least 10% across the economic cycle. Extremely strong vertical supply chains and long-term contractual off-take. Extremely strong and publicly articulated risk management in place with superior track record. No proprietary trading in commodities or derivatives.	GTC: Highly diversified global operations. No concentrations in geography, end-market or customer. Expected to have very stable and high earnings evidenced by ROA of at least 6% across the economic cycle. Very strong vertical supply chains and long-term contractual off-take. Very strong and publicly articulated risk management in place with superior track record. No proprietary trading in commodities or derivatives.	GTC: Highly diversified global operations. Multiple business segments and a wide range of service in all segments. Some concentration in geography, end market or customer. Expected to have stable and very strong earnings with very low volatility across the economic cycle. Strong vertical supply chains and long-term contractual off-take. Strong and publicly articulated risk management in place with good track record. No proprietary trading in commodities or derivatives.	GTC: Global operations with strong diversification. Multiple business segments and a wide range of service in most segments. Moderate concentration in geography, end market or customer. Expected to have robust earnings with low volatility. Expected to have degree of volatility in cash flow across the cycle but capable of being offset by asset recycling.[4] Very good vertical supply chains and long-term contractual off-take. Strong risk management in place with good track record. Minimal proprietary trading in commodities. No proprietary trading in derivatives.	GTC: Broad continental[5] presence with some regional presence. Several business segments with broad service offerings in many segments. Moderate concentration in geography, end market or customer. Moderate expected volatility in results. Expected to have volatility in cash flow across the cycle but capable of being offset by asset recycling. Good vertical supply chains, some long term contractual off-take. Good risk management in place with good track record. Minimal proprietary trading in commodities. No proprietary trading in derivatives.	GTC: Broad continental presence. Operates in a few business segments with broad service offerings in at least one key segment. Significant concentration in geography, end market or customer. Expected to have volatility in earnings and cash flow across the cycle but capable of being offset by asset recycling. Some vertical supply chains and degree of long-term off-take. Sufficient risk management in place with adequate track record. Small proprietary trading in commodities. No proprietary trading in derivatives.	GTC: Regional operator within a single country or continent. Substantial concentrations in geography, end market or customer. Expected to have high volatility in cash flow across the cycle. Very limited potential for flexibility in asset recycling. Minimal vertical integration or long-term contractual off-take. Inadequate risk management with minimal track record. Material proprietary trading in commodities. Small proprietary trading in derivatives.	GTC: Wholly localized and concentrated operations within one country. No vertical integration or long-term contracts. Expected to have extremely high volatility in both earnings and cash flow. Inadequate risk management with poor track record. Material proprietary trading in commodities. Material proprietary trading in derivatives.

OUTDATED METHODOLOGY

	GTC Weight	CTC Weight	Aaa	Aa	A	Baa	Ba	B	Caa	Ca
Business Profile	0%	30%	CTC: No merchandising activities. All trading tied to vertically integrated businesses. Geographically diverse sales. Long history of an extremely diverse product portfolio. Number one global market shares across all businesses as evidenced by operating margins of at least 15% on a sustainable basis. Competitors are significantly smaller in size. Use of derivatives tied directly to physical positions and logistical assets. No proprietary trading.	CTC: Minimal merchandising activities. Geographically diverse sales. Long history of an extremely diverse product portfolio. Leading global market shares across all businesses as evidenced by operating margins of at least 10% on a sustainable basis. Competitors are significantly smaller in size. Use of derivatives tied directly to physical positions and logistical assets. No proprietary trading.	CTC: More significant merchandising operations but largely supporting vertically integrated businesses. Largest region accounts for less than 40% of total sales. Long history of a diverse product portfolio. Leading global market share in all major product lines. Few competitors of similar size & many smaller competitors. Vast majority of derivatives tied directly to physical positions and logistical assets. Minimal use of proprietary trading. Management policies prevent individuals from incurring proprietary trading risk. Compensation policies do not provide any benefit to individuals for incurring proprietary trading risk.	CTC: Majority of business is tied to merchandising and has a good global logistics infrastructure. Largest region accounts for less than 50% of total sales. Long established market positions across a moderately diverse product portfolio. Leading global market share in key segments. Few larger competitors & many smaller competitors. Derivative primarily used to support physical positions or logistical assets. Proprietary trading is tightly controlled by established policies and tight limits for all individuals. Management is unable to expand risk limits without Board approval. Compensation policies discourage individuals from incurring proprietary trading risk.	CTC: Majority of business is tied to merchandising and it has adequate logistics assets. Largest region accounts for less than 65% of total sales. Moderately diverse product portfolio. Established & stable market positions globally. Many larger competitors & many competitors of similar size. Derivatives largely used to support physical positions or logistical assets. Proprietary trading is controlled by established policies and strict risk limits. Management has limited ability to expand risk limits without Board approval. Compensation policies provide limited incentives for proprietary trading risk.	CTC: Merchandising is largest business and has good regional logistics assets. Largest region accounts for less than 90% of total sales. Limited product portfolio. Established & stable market positions in more than one region. Significantly larger competitors and many competitors of similar size. Substantial use of derivatives to support merchandising/trading that is not tied to the company's logistical assets. Proprietary trading is controlled by established policies and risk limits. Management has some discretion over risk limits (or risk limits are set at elevated levels). Compensation policies provide incentives for traders to take risk, but discourage excessive risk.	CTC: Merchandising is largest business and has limited regional logistics assets. Largest region accounts for less than 95% of total sales. Two or three main products. Established & stable market position in one region. Most competitors are significantly larger and few competitors of similar size. Majority of derivatives used to support merchandising / trading that is not tied to the company's logistical assets. Proprietary trading is controlled by established policies and risk limits. Management has substantial discretion over risk limits (or risk limits are set at excessive levels). Compensation policies do not discourage traders from taking excessive risk.	CTC: Merchandising is largest business but has no logistics assets. Largest region accounts for ≥ 95% of total sales. One main product. Established but volatile market position in one region. Smallest company in market or region. Vast majority of derivatives used to support merchandising / trading that is not tied to the company's logistical assets. Proprietary trading is not controlled by established policies and risk limits. Management has substantial discretion over risk limits (or risk limits are set at excessive levels). Compensation policies do not discourage traders from taking excessive risk.

OUTDATED METHODOLOGY

	GTC Weight	CTC Weight	Aaa	Aa	A	Baa	Ba	B	Caa	Ca
Factor: Leverage and Coverage (20%)										
Debt / EBITDA[6]	10%	0%	≤ 1.5x	1.5x - 2.5x	2.5x - 3.5x	3.5x - 4.5x	4.5x - 6x	6x - 7.5x	7.5x - 9x	> 9x
Debt / EBITDA[6]	0%	10%	≤ 0.5x	0.5x - 1x	1x - 2x	2x - 3x	3x - 4.5x	4.5x - 6.5x	6.5x - 8.5x	> 8.5x
RCF / Net Debt[7]	10%	0%	≥ 50%	35% - 50%	25% - 35%	15% - 25%	7% - 15%	0% - 7%	(5)% - 0%	< (5)%
RCF / Net Debt[8]	0%	10%	≥ 70%	50% - 70%	35% - 50%	20% - 35%	12.5% - 20%	5% - 12.5%	0% - 5%	< 0%
Factor: Financial Policy (30%)										
Financial Policy	30%		Expected to have extremely conservative financial policies (including risk and liquidity management); very stable metrics; essentially no event risk that would cause a rating transition; and public commitment to a very strong credit profile over the long term.	Expected to have very conservative financial policies (including risk and liquidity management); stable metrics; minimal event risk that would cause a rating transition; and public commitment to a strong credit profile over the long term.	Expected to have predictable financial policies (including risk and liquidity management) that preserve creditor interests; although modest event risk exists, the effect on leverage is likely to be small and temporary; strong commitment to a solid credit profile.	Expected to have financial policies (including risk and liquidity management) that balance the interests of creditors and shareholders; some risk that debt-funded acquisitions or shareholder distributions could lead to a weaker credit profile.	Expected to have financial policies (including risk and liquidity management) that tend to favor shareholders over creditors; above-average financial risk resulting from shareholder distributions, acquisitions or other significant capital structure changes.	Expected to have financial policies (including risk and liquidity management) that favor shareholders over creditors; high financial risk resulting from shareholder distributions, acquisitions or other significant capital structure changes.	Expected to have financial policies (including risk and liquidity management) that create elevated risk of debt restructuring in varied economic environments.	Expected to have financial policies (including risk and liquidity management) that create elevated risk of debt restructuring even in healthy economic environments.

OUTDATED METHODOLOGY

[1] For the linear scoring scale, the Aaa endpoint value is $300 billion. A value of $300 billion or better equates to a numeric score of 0.5. The Ca endpoint value is zero. A value of zero equates to a numeric score of 20.5.
[2] For the linear scoring scale, the Aaa endpoint value is $250 billion. A value of $250 billion or better equates to a numeric score of 0.5. The Ca endpoint value is zero. A value of zero equates to a numeric score of 20.5.
[3] For the linear scoring scale, the Aaa endpoint value is $100 billion. A value of $100 billion or better equates to a numeric score of 0.5. The Ca endpoint value is zero. A value of zero equates to a numeric score of 20.5.
[4] The term asset recycling in this context refers to the process of general trading companies (and some others) routinely investing and subsequently divesting assets. This regular turnover may involve large scale assets that are non-core to the company's operations but such activities form an ongoing part of the company's expected business operations. In times of financial stress, for example, a trading company which has diverse assets in numerous sectors may choose to liquidate one or several assets in order to alleviate some aspects of its financial position, for example high leverage or weak liquidity.
[5] The term continental refers to the American, European and Asian continents.
[6] For the linear scoring scale, the Aaa endpoint value is 0x. A value of 0x equates to a numeric score of 0.5. The Ca endpoint value is 12x. A value of 12x or worse equates to a numeric score of 20.5, as does a negative debt/EBITDA value.
[7] For the linear scoring scale, when net debt is positive, the Aaa endpoint value is 75%. A value of 75% or better equates to a numeric score of 0.5. The Ca endpoint value is (10)%. A value of (10)% or worse equates to a numeric score of 20.5. When net debt is negative and RCF is positive, the numeric score is 0.5. When net debt is negative and RCF is negative or zero, the numeric score is 16.5.
[8] For the linear scoring scale, when net debt is positive, the Aaa endpoint value is 100%. A value of 100% or better equates to a numeric score of 0.5. The Ca endpoint value is (5)%. A value of (5)% or worse equates to a numeric score of 20.5. When net debt is negative and RCF is positive, the numeric score is 0.5. When net debt is negative and RCF is negative or zero, the numeric score is 19.5.
Source: Moody's Ratings

Discussion of the scorecard factors

In this section, we explain our general approach for scoring each scorecard factor or sub-factor, and we describe why they are meaningful as credit indicators.

Factor: Scale (20% weight)

Why it matters

Scale is an important indicator of the overall depth of a company's business and its success in attracting a variety of customers, as well as its resilience to shocks, such as sudden shifts in demand or rapid cost increases. Large revenues and assets are typically indicative of a more sustainable business position that enables a company to have a greater influence over business trends and pricing, and to weather the vagaries of economic cycles and support a stable or growing market position. Scale also can be an indicator of greater resilience to changes in product demand, geographic diversity, cost absorption and bargaining strength with customers and suppliers.

This factor comprises two sub-factors: Revenue and Assets.

Revenue

Size in this sector implies ability to benefit from economies of scale in sourcing and distribution of products on a preferred basis. Size also tends to be indicative of a more diversified portfolio of commodities and vertically integrated businesses.

Total Assets (General Trading Companies)

General trading companies are involved in trading businesses that result in highly liquid trade receivables and inventories that may represent a material portion of the total balance sheet. These companies are also often involved in long-term, illiquid business investments along with their trading businesses. Higher levels of total assets tend to be associated with a more diversified portfolio, large operating franchises, as well as the potential for larger cash generation through divestments.

Fixed Assets (Commodity Trading Companies)

Fixed assets is another indicator of scale. Higher levels of fixed assets investment tend to be consistent with a more stable performance for commodity trading companies, including earnings and cash flow performance. Fixed assets is a relatively stable indicator of scale compared to total assets, which is impacted by seasonal volume and commodity price changes.

How we assess it for the scorecard

Scoring for this factor is based on two sub-factors: Revenue and Assets.

In assessing assets, we use different scoring criteria for general trading companies and commodity trading companies, in recognition of differences in their respective business models.

Revenue

Scale is measured (or estimated in the case of forward-looking expectations) using total reported revenue in billions of US dollars.

Total Assets (General Trading Companies)

We use total assets balance sheet data for the latest fiscal year-end.

Fixed Assets (Commodity Trading Companies)

We use unadjusted gross property, plant and equipment (PP&E). We did not choose to use net PP&E in the scorecard because most merchandising assets are depreciated well before the end of their useful life. Therefore, in most cases the gross number best reflects the level of assets in service.

Factor: Business Profile (30% weight)

Why it matters

The business profile of a trading company is important because it greatly influences its ability to generate sustainable earnings and operating cash flows. It provides important indications of the strength of the company's global presence, the diversity of its products, its long-term competitiveness, the stability of its performance, its long-term viability and its risk profile. Our assessment includes:

OUTDATED METHODOLOGY

geographic, operational and product diversity; competitive advantages; durability of its market share and customer relationships; the stability of performance over time; the competitive landscape in each key market; the threat posed by potential new entrants or technological change; the degree to which products or services are differentiated; growth strategy; and risk profile. The company's risk profile considers a broad range of issues, including the perceived likelihood that the company might undertake sizable or frequent acquisitions in new markets that would raise business risk, exposures to volatile commodity prices, and management's policies and practices concerning proprietary trading. For commodity trading companies, we also consider the level of vertical integration, the percentage of sales and earnings that arise from merchandising activities and the evolution of the company's business over time.

How we assess it for the scorecard

Scoring for this factor is based on our qualitative assessment of each company, including its overall market position, product, operational and geographic diversity, expected growth prospects and stability of cash flows.

We generally do not expect a given company's business profile to exactly match each of the attributes for a given scoring category. We typically assign the factor score based on the alpha category for which the company has the greatest number of characteristics. However, there may be cases in which one characteristic, such as the company's risk profile, competitive position or stability of performance, is sufficiently important to a particular company's credit profile that it has a large influence on the factor score. We use different scoring criteria for this factor for general trading companies and commodity trading companies, in recognition of the differences in their respective business models and the resulting differences in their competitive landscapes.

General trading companies: We view general trading companies that are more highly diversified and that have strong vertical supply chains, in which they can add material value at successive stages of the production chain (not simply conveyance or logistics), as those best equipped to maintain a strong business model with resilient and high earnings and low volatility through the economic cycle. Companies that score low on this factor are those that are not globally diverse, have little or no ability to add value within a vertical supply chain and are perceived as having greater vulnerability to adverse consequences in areas that include proprietary trading and use of derivatives.

Commodity trading companies: Our qualitative assessment of a commodity trading company's business profile considers the nature of risks related to business activities, the competitive landscape, the diversity of the business, market shares, as well as other considerations. Our risk assessment considers business activities, including merchandising, proprietary trading positions, policies, and controls. Additionally, management's track record of adhering to its policies, changing those policies when appropriate and proactively addressing potential future concerns is important in this assessment. Our assessment of a company's risk profile also incorporates the evolution over time of various characteristics, including its risk management systems and policies, as well as our view of the level and quality of disclosure.

We consider a number of aspects in assessing the competitive landscape with particular emphasis on diversity, the nature of competition, companies with distinct competitive advantages and the economic benefit these advantages provide. We assess these characteristics partly by evaluating companies relative to their direct competitors.

We consider the number of significant and distinct business segments, the range of products or services offered, and end market and customer diversity. Companies with multiple business segments and a wide range of products and services tend to exhibit greater stability in operating results when compared to competitors with a narrower business focus. Conversely, companies that serve only one market may be more vulnerable to competitive pressures, or other exogenous events, thereby experiencing greater volatility in earnings and cash flows. Geographic diversity is also important, as a company with a narrow or regional focus can be affected negatively by regional economic events, government policies and the weather, whereas such risk is mitigated in companies with offerings that span many regions.

Strong market share in a company's key markets suggests a sustainable business position with greater ability to weather volatile market conditions. Market share that is protected by the size or characteristics of logistics assets, government regulations, contractual agreements or unique licensing restrictions can underpin a high market share. A company with high market shares across its major businesses can typically generate higher margins and returns than its competitors.

OUTDATED METHODOLOGY

Factor: Leverage and Coverage (20% weight)

Why it matters

Leverage and coverage measures provide important indications of the financial flexibility and long-term viability of a company, including its ability to adapt to changes in the economic and business environments in the segments in which it operates.

This factor comprises two quantitative sub-factors:

Debt / EBITDA

The ratio of total debt to earnings before interest, taxes, depreciation and amortization (Debt/EBITDA) is an indicator of debt serviceability and financial leverage. The ratio is commonly used in this sector as a proxy for comparative financial strength.

RCF / Net Debt

The ratio of retained cash flow to net debt (RCF/Net Debt) is an indicator of a company's cash generation (before working capital movements and capital expenditures, and after dividend payments) relative to its net debt.

How we assess it for the scorecard

Scoring for this factor is based on two sub-factors: Debt/EBITDA and RCF/Net Debt.[1]

Debt / EBITDA

The numerator is total debt, and the denominator is EBITDA.

General trading companies: The EBITDA for general trading companies normally includes equity income as these investments are part of their fundamental business operations. This tends to be volatile as equity income can fluctuate with investments and divestments. Generally, the companies are highly levered, due to the nature of the finance function performed by general trading companies, such as engaging in leasing and captive-finance-related businesses, as well as providing the short-term financing associated with their trading businesses. General trading companies are engaged in finance-related businesses such as providing short-term financing to their customers, as well as engaging in leasing and captive-finance-related businesses. These attributes give general trading companies a financial profile – and leverage – more akin to a hybrid between a corporate and a financial institution.

Commodity trading companies: While this metric varies greatly depending on commodity prices, we expect companies will target an average range over time, irrespective of commodity prices. In addition, we may reduce debt to reflect a certain amount of Readily Marketable Inventories (RMI), which can be monetized to reduce leverage. The level of RMI is determined for each company based on its product and business mix (for more information, see the appendix).

RCF / Net Debt

The numerator is retained cash flow, and the denominator is net debt.

General trading companies: We may make adjustments to gross/net debt to exclude captive-finance-related obligations. However, these companies' businesses are complex and intertwined, and captive-finance-related debt is often not separately disclosed.

Commodity trading companies: We make several adjustments that address specific characteristics of this industry sector (e.g., readily marketable inventory and time charters). Please see the appendix.

Factor: Financial Policy (30% weight)

Why it matters

Financial policy encompasses management and board tolerance for financial risk and commitment to a strong credit profile. It is an important rating determinant, because it directly affects debt levels, credit quality, the future direction for the company and the risk of adverse changes in financing and capital structure.

Financial risk tolerance serves as a guidepost to investment and capital allocation. An expectation that management will be committed to sustaining an improved credit profile is often necessary to support an upgrade. For example, we may not upgrade the ratings of a company that has built flexibility within its rating category if we believe the company will use that flexibility to fund a strategic

OUTDATED METHODOLOGY

acquisition, cash distribution to shareholders, spinoff or other leveraging transaction. Conversely, a company's credit rating may be better able to withstand a moderate leveraging event if management places a high priority on returning credit metrics to pre-transaction levels and has consistently demonstrated the commitment to do so through prior actions. Liquidity management is an important aspect of overall risk management and can provide insight into risk tolerance. Liquidity management is distinct from the level of liquidity, which is discussed in the "Other considerations" section.

General trading companies have historically frequently used acquisitions as a continuous and ongoing feature of their unique business model, in order to spur revenue growth, expand business lines, consolidate market positions, advance cost synergies, seek to access new technology, or move into or out of (in the latter case, via divestments) dynamic opportunities that they may identify across the globe due to their broad footprints and diverse exposures. The impact of an acquisition on a rating will partly depend on the company's existing capital structure and the degree to which it is changed by the acquisition.

How we assess it for the scorecard

We assess the issuer's desired capital structure or targeted credit profile, its history of prior actions, including its track record of risk and liquidity management, and its adherence to its commitments. Attention is paid to management's operating performance and use of cash flow through different phases of economic and industry cycles. Also of interest is the way in which management responds to key events, such as changes in the credit markets and liquidity environment, legal actions, competitive challenges or regulatory pressures. Considerations include a company's public commitments in this area, its track record of adhering to commitments and our views on the ability of the company to achieve its targets.

When considering event risks in the context of scoring financial policy, we assess the likelihood and potential negative impact of M&A or other types of balance-sheet-transforming events. Management's appetite for M&A activity is assessed, with a focus on the type of transactions (i.e., core competency or new business) and funding decisions. Frequency and materiality of acquisitions and previous financing choices are evaluated. A history of debt-financed or credit-transforming acquisitions generally results in a lower score for this factor. We may also consider negative repercussions caused by shareholders' willingness to sell the company.

We also consider a company's and its owners' past record of balancing shareholder returns and debtholders' interests. A track record of favoring shareholder returns at the expense of debtholders is likely to be viewed negatively in scoring this factor.

Other considerations

Ratings may reflect consideration of additional factors that are not in the scorecard, usually because the factor's credit importance varies widely among the issuers in the sector or because the factor may be important only under certain circumstances or for a subset of issuers. Such factors can include financial controls and the quality of financial reporting; corporate legal structure; the quality and experience of management; assessments of corporate governance, as well as environmental and social considerations; exposure to uncertain licensing regimes; and possible government interference in some countries. Regulatory, litigation, liquidity, technology and reputational risk, as well as changes to consumer and business spending patterns, competitor strategies and macroeconomic trends, can also affect ratings.

The following are examples of additional considerations that may be reflected in our ratings and that may cause ratings to be different from scorecard-indicated outcomes.

Environmental, social and governance considerations

Where environmental, social and governance (ESG) issues are meaningful for credit profiles, we incorporate them into our ratings analysis in a variety of ways in the application of our sector-specific methodologies. As one part of our overall credit analysis, we consider how ESG considerations could affect the qualitative and quantitative factors and sub-factors in the scorecard.

Even where ESG considerations do not affect the measures in the scorecard, or where they cannot be quantified, we incorporate them into our overall analysis of credit drivers that are meaningful to the rating. As a result, we may incorporate these ESG risks qualitatively outside of the scorecard. As part of our ratings analysis, we may establish Issuer Profile Scores (IPSs), which indicate our opinion of the extent to which a given issuer is exposed to E, S and G risks (incorporating ESG-specific mitigants) or benefits from its exposure to E, S or G. The IPSs are inputs to credit ratings. For more information, please see our methodology that describes our general principles for assessing ESG risks.[2]

OUTDATED METHODOLOGY

Liquidity

Liquidity is an important rating consideration for all trading companies, and extremely weak liquidity can heavily affect ratings in many cases. However, the relative strength or weakness of liquidity may not have a substantial impact in discriminating between two issuers with an otherwise similar credit profile. Liquidity can be particularly important where companies have large short-term demands on liquidity, for example in highly seasonal operating environments where working capital needs must be considered. We form an opinion on likely near-term liquidity requirements from the perspective of both sources and uses of cash. For more information on our approach, please see our liquidity cross-sector methodology.[3]

Financial controls

We rely on the accuracy of audited financial statements to assign and monitor ratings in this sector. The quality of financial statements may be influenced by internal controls, including the proper tone at the top, centralized operations and consistency in accounting policies and procedures. Auditors' reports on the effectiveness of internal controls, auditors' comments in financial reports and unusual restatements of financial statements or delays in regulatory filings may indicate weaknesses in internal controls.

Management strategy

The quality of management is an important factor supporting a company's credit strength. Assessing the execution of business plans over time can be helpful in assessing management's business strategies, policies and philosophies and in evaluating management performance relative to performance of competitors and our projections. Management's track record of adhering to stated plans, commitments and guidelines provides insight into management's likely future performance, including in stressed situations.

Additional metrics

The metrics included in the scorecard are those that are generally most important in assigning ratings to companies in this industry; however, we may use additional metrics to inform our analysis of specific companies. These additional metrics may be important to our forward view of metrics that are in the scorecard or other rating factors.

Aside from the financial ratios included in the scorecard, other financial measures can provide useful indications of a company's relative credit strength and how this changes over time. For example, Net Debt/Net Working Capital provides an indication of how much of the balance sheet debt is financing relatively liquid assets and can be a useful credit indicator in some cases for commodity trading companies. For most pure commodity merchandising/trading companies, this metric is below 1.0x. For the large, vertically integrated companies, this ratio can be between 1x-2x as they have a much larger investment in fixed assets or affiliate investments. These typical ranges can change when commodity prices are at peak or trough levels.

Event risk

We also recognize the possibility that an unexpected event could cause a sudden and sharp decline in an issuer's fundamental creditworthiness, which may cause actual ratings to be lower than the scorecard-indicated outcome. Event risks — which are varied and can range from leveraged recapitalizations to sudden regulatory changes or liabilities from an accident — can overwhelm even a stable, well-capitalized firm. Some other types of event risks include M&A, asset sales, spinoffs, shareholder distributions, litigation, pandemics, significant cybercrime events and geopolitical conflicts.

Parental support

Ownership can provide ratings lift for a particular company in the trading companies sector if it is owned by a highly rated owner(s) and is viewed to be of strategic importance to those owners. In our analysis of parental support, we consider whether the parent has the financial capacity and strategic incentives to provide support to the issuer in times of stress or financial need (e.g., a major capital investment or advantaged operating agreement), or has already done so in the past. Conversely, if the parent puts a high dividend burden on the issuer, which in turn reduces its flexibility, the ratings would reflect this risk.

Government-related issuers may receive ratings uplift due to expected government support. However, for certain issuers, government ownership can have a negative impact on the underlying Baseline Credit Assessment.[4] For example, price controls, onerous taxation and high distributions can have a negative effect on an issuer's underlying credit profile.

OUTDATED METHODOLOGY

Other institutional support

In some countries, large corporate issuers have received government or banking support in the event of financial difficulties because of their overall importance to the functioning of the economy. In Japan, our corporate ratings consider the support that has operated there for large and systemically important organizations. Over the years, this has resulted in lower levels of default than might otherwise have occurred. Our approach considers whether the presence of group and banking relationships may provide support when systemically important companies encounter significant financial stress.

Using the scorecard to arrive at a scorecard-indicated outcome

1. Measurement or estimation of factors in the scorecard

In the "Discussion of the scorecard factors" section, we explain our analytical approach for scoring each scorecard factor or sub-factor, and we describe why they are meaningful as credit indicators. When a factor comprises sub-factors, we score at the sub-factor level. Some factors do not have sub-factors, in which case we score at the factor level.

The information used in assessing the sub-factors is generally found in or calculated from information in the company's financial statements or regulatory filings, derived from other observations or estimated by Moody's analysts. We may also incorporate non-public information.

Our ratings are forward-looking and reflect our expectations for future financial and operating performance. However, historical results are helpful in understanding patterns and trends of a company's performance as well as for peer comparisons. Financial metrics,[5] unless otherwise indicated, are typically calculated based on an annual or other 12-month period. However, the factors in the scorecard can be assessed using various time periods. For example, rating committees may find it analytically useful to examine both historical and expected future performance for periods of several years or more.

All of the quantitative credit metrics incorporate our standard adjustments[6] to income statement, cash flow statement and balance sheet amounts for items such as underfunded pension obligations and operating leases. We may also make other analytical adjustments that are specific to a particular company.

2. Mapping scorecard factors to a numeric score

After estimating or calculating each factor or sub-factor, each outcome is mapped to a broad Moody's rating category (Aaa, Aa, A, Baa, Ba, B, Caa, or Ca, also called alpha categories) and to a numeric score.

Qualitative factors are scored based on the description by broad rating category in the scorecard. The numeric value of each alpha score is based on the scale below.

Exhibit 3

Numeric equivalents for qualitative factor and sub-factor scores

Aaa	Aa	A	Baa	Ba	B	Caa	Ca
1	3	6	9	12	15	18	20

Source: Moody's Ratings

Quantitative factors are scored on a linear continuum. For each metric, the scorecard shows the range by alpha category. We use the scale below and linear interpolation to convert the metric, based on its placement within the scorecard range, to a numeric score, which may be a fraction. As a purely theoretical example, if there were a ratio of revenue to interest for which the Baa range was 50x to 100x, then the numeric score for an issuer with revenue/interest of 99x, relatively strong within this range, would score closer to 7.5, and an issuer with revenue/interest of 51x, relatively weak within this range, would score closer to 10.5. In the scorecard footnotes, we define the endpoints of the line (i.e., the value of the metric that constitutes the lowest possible numeric score, and the value that constitutes the highest possible numeric score).

OUTDATED METHODOLOGY

Exhibit 4
Ranges of numeric equivalents for quantitative factor and sub-factor scores

Aaa	Aa	A	Baa	Ba	B	Caa	Ca
0.5 - 1.5	1.5 - 4.5	4.5 - 7.5	7.5 - 10.5	10.5 - 13.5	13.5 - 16.5	16.5 - 19.5	19.5 - 20.5

Source: Moody's Ratings

3. Determining the overall scorecard-indicated outcome

The numeric score for each sub-factor (or each factor, when the factor has no sub-factors) is multiplied by the weight for that sub-factor (or factor), with the results then summed to produce an aggregate numeric score. The aggregate numeric score is then mapped back to a scorecard-indicated outcome based on the ranges in the table below.

Exhibit 5
Scorecard-indicated outcome

Aggregate numeric score	Scorecard-indicated outcome
x ≤ 1.5	Aaa
1.5 < x ≤ 2.5	Aa1
2.5 < x ≤ 3.5	Aa2
3.5 < x ≤ 4.5	Aa3
4.5 < x ≤ 5.5	A1
5.5 < x ≤ 6.5	A2
6.5 < x ≤ 7.5	A3
7.5 < x ≤ 8.5	Baa1
8.5 < x ≤ 9.5	Baa2
9.5 < x ≤ 10.5	Baa3
10.5 < x ≤ 11.5	Ba1
11.5 < x ≤ 12.5	Ba2
12.5 < x ≤ 13.5	Ba3
13.5 < x ≤ 14.5	B1
14.5 < x ≤ 15.5	B2
15.5 < x ≤ 16.5	B3
16.5 < x ≤ 17.5	Caa1
17.5 < x ≤ 18.5	Caa2
18.5 < x ≤ 19.5	Caa3
19.5 < x ≤ 20.5	Ca
x > 20.5	C

Source: Moody's Ratings

OUTDATED METHODOLOGY

For example, an issuer with an aggregate numeric score of 11.7 would have a Ba2 scorecard-indicated outcome.

In general, the scorecard-indicated outcome is oriented to the corporate family rating (CFR) for speculative-grade issuers and to the senior unsecured rating for investment-grade issuers. For issuers that benefit from ratings uplift from parental support, government ownership or other institutional support, we consider the underlying credit strength or Baseline Credit Assessment for comparison to the scorecard-indicated outcome. For an explanation of the Baseline Credit Assessment, please refer to *Rating Symbols and Definitions* and to our cross-sector methodology for government-related issuers.[7]

Assigning issuer-level and instrument-level ratings

After considering the scorecard-indicated outcome, other considerations and relevant cross-sector methodologies, we typically assign a corporate family rating (CFR) to speculative-grade issuers or a senior unsecured rating for investment-grade issuers. For issuers that benefit from rating uplift from government ownership, we may assign a Baseline Credit Assessment.[8]

Individual debt instrument ratings may be notched up or down from the CFR or the senior unsecured rating to reflect our assessment of differences in expected loss related to an instrument's seniority level and collateral. The documents that provide broad guidance for such notching decisions are the rating methodology on loss given default for speculative-grade nonfinancial companies, the methodology for notching corporate instrument ratings based on differences in security and priority of claim, and the methodology for assigning short-term ratings.[9]

Key rating assumptions

For information about key rating assumptions that apply to methodologies generally, please see *Rating Symbols and Definitions*.[10]

Limitations

In the preceding sections, we have discussed the scorecard factors and many of the other considerations that may be important in assigning ratings. In this section, we discuss limitations that pertain to the scorecard and to the overall rating methodology.

Limitations of the scorecard

There are various reasons why scorecard-indicated outcomes may not map closely to actual ratings.

The scorecard in this rating methodology is a relatively simple reference tool that can be used in most cases to approximate credit profiles of companies in this sector and to explain, in summary form, many of the factors that are generally most important in assigning ratings to these companies. Credit loss and recovery considerations, which are typically more important as an issuer gets closer to default, may not be fully captured in the scorecard. The scorecard is also limited by its upper and lower bounds, causing scorecard-indicated outcomes to be less likely to align with ratings for issuers at the upper and lower ends of the rating scale.

The weights for each factor and sub-factor in the scorecard represent an approximation of their importance for rating decisions across the sector, but the actual importance of a particular factor may vary substantially based on an individual company's circumstances.

Factors that are outside of the scorecard, including those discussed above in the "Other considerations" section, may be important for ratings, and their relative importance may also vary from company to company. In addition, certain broad methodological considerations described in one or more cross-sector methodologies may be relevant to ratings in this sector.[11] Examples of such considerations include the following: how sovereign credit quality affects non-sovereign issuers, the assessment of credit support from other entities, the relative ranking of different classes of debt and hybrid securities, and the assignment of short-term ratings.

We may use the scorecard over various historical or forward-looking time periods. Furthermore, in our ratings we often incorporate directional views of risks and mitigants in a qualitative way.

General limitations of the methodology

This methodology document does not include an exhaustive description of all factors that we may consider in assigning ratings in this sector. Companies in the sector may face new risks or new combinations of risks, and they may develop new strategies to mitigate risk. We seek to incorporate all material credit considerations in ratings and to take the most forward-looking perspective that visibility into these risks and mitigants permits.

Ratings reflect our expectations for an issuer's future performance; however, as the forward horizon lengthens, uncertainty increases and the utility of precise estimates, as scorecard inputs or in other considerations, typically diminishes. Our forward-looking opinions are based on assumptions that may prove, in hindsight, to have been incorrect. Reasons for this could include unanticipated changes in any of the following: the macroeconomic environment, general financial market conditions, industry competition, disruptive technology, or regulatory and legal actions. In any case, predicting the future is subject to substantial uncertainty.

OUTDATED METHODOLOGY

Appendix: Additional financial adjustments for commodity trading companies

In addition to our standard analytical adjustments to financial statements, we may make additional adjustments to the financial statements of commodity trading companies. This appendix provides insights on adjustments we make to improve the comparability of data and provide more meaningful financial metrics.

Time charters: The leasing of vessels for the transport of commodities is a necessary part of this business. Ready access to suitable ships allows companies to take advantage of short-term price inequalities in the market and generate higher profits in the logistics/distribution business. However, due to the nature of the commodity trading business, a firm cannot utilize 100% of its ships' capacity for its own products and typically seeks to lease out the unused capacity. Some firms manage their exposure by hedging their projected requirements for the following periods as there is a very liquid futures market for time charters. Other firms have chosen to become brokers – leasing a much greater number of ships than they can utilize, and sub-leasing the vast majority of their shipping capacity and making this a profit center. To the extent that a company has a meaningful amount of time charters that do not meet the accounting definition of a lease and thus does not capitalize them in accordance with US GAAP or IFRS, we would make adjustments similar to our treatment of leases.[12]

Readily marketable inventories (RMI): The vast majority of the inventory held by these companies is typically either fully hedged or contractually sold and the turnover of inventory is fairly rapid (excluding inventories specifically held to take advantage of the futures versus cash price differential). This adjustment is typically not suitable for other commodity industries as they do not have their entire inventory hedged or contractually sold. Accordingly, companies could in theory liquidate a portion of their inventory to reduce debt. However, we do not believe that companies can monetize a large percentage of their RMI without adversely impacting their business – reducing future profitability or raising concerns among trading partners or counterparties (potential for increased collateral requirements in OTC trades). Hence, we may view a smaller portion of RMI as benefiting liquidity than total reported RMI.

For trading companies, commodity price volatility creates even greater volatility in financial metrics based on net debt. We have found that using a conservative value for RMI as a deduction to debt provides some stability to the companies' net debt financial metrics, enabling us to communicate a reasonable range for these financial metrics where the rating of the company would likely remain stable, as well as the values for these metrics that could cause us to move the rating for the company up or down. We do not use the RMI numbers generated by the companies as they would provide too large of an adjustment to net debt, allowing debt to increase substantially without a meaningful change in these financial metrics.

We make estimated adjustments for RMI that are typically between 20%-50% of total inventory depending on the proportion of the company's revenue base and inventory that is tied to their merchandising/trading operations. On a case-by-case basis and depending on the specific circumstances and market outlook for a particular commodity or basket of commodities, a higher percentage adjustment could be contemplated (than the 20%-50% range cited above), if we have sufficient insight into the specific risks. Generally, such an adjustment would be no higher than 75% of the inventory concerned. The RMI adjustment is subtracted from debt in the calculation of commodity trading companies' Debt/EBITDA and RCF/Net Debt metrics in this methodology.

OUTDATED METHODOLOGY

Moody's related publications

Credit ratings are primarily determined through the application of sector credit rating methodologies. Certain broad methodological considerations (described in one or more cross-sector methodologies) may also be relevant to the determination of credit ratings of issuers and instruments. A list of sector and cross-sector methodologies can be found here.

For data summarizing the historical robustness and predictive power of credit ratings, please click here.

For further information, please refer to *Rating Symbols and Definitions*, which is available here.

OUTDATED METHODOLOGY

Endnotes

1 For definitions of our most common metrics, please see our cross-sector methodology that describes our financial statement adjustments in the analysis of nonfinancial corporations. A link to a list of our sector and cross-sector methodologies can be found in the "Moody's related publications" section.

2 A link to a list of our sector and cross-sector methodologies can be found in the "Moody's related publications" section.

3 A link to a list of our sector and cross-sector methodologies can be found in "Moody's related publications" section.

4 For an explanation of the Baseline Credit Assessment, please refer to *Rating Symbols and Definitions* and to our cross-sector methodology for government-related issuers. A link to *Rating Symbols and Definitions* and a link to a list of our sector and cross-sector methodologies can be found in the "Moody's related publications" section.

5 For definitions of our most common metrics, please see our cross-sector methodology that describes our financial statement adjustments in the analysis of nonfinancial corporations. A link to a list of our sector and cross-sector methodologies can be found in the "Moody's related publications" section.

6 For an explanation of our standard adjustments, please see our cross-sector methodology that describes our financial statement adjustments in the analysis of nonfinancial corporations. A link to a list of our sector and cross-sector methodologies can be found in the "Moody's related publications" section.

7 A link to *Rating Symbols and Definitions* and a link to a list of our sector and cross-sector methodologies can be found in the "Moody's related publications" section.

8 For an explanation of the Baseline Credit Assessment, please refer to *Rating Symbols and Definitions* and to our cross-sector methodology for government-related issuers. A link to *Rating Symbols and Definitions* and a link to a list of our sector and cross-sector methodologies can be found in the "Moody's related publications" section.

9 A link to a list of our sector and cross-sector methodologies can be found in the "Moody's related publications" section.

10 A link to *Rating Symbols and Definitions* can be found in the "Moody's related publications" section.

11 A link to a list of our sector and cross-sector methodologies can be found in the "Moody's related publications" section.

12 A link to a list of our sector and cross-sector methodologies can be found in the "Moody's related publications" section.

OUTDATED METHODOLOGY

© 2025 Moody's Corporation, Moody's Investors Service, Inc., Moody's Analytics, Inc. and/or their licensors and affiliates (collectively, "MOODY'S"). All rights reserved.
CREDIT RATINGS ISSUED BY MOODY'S CREDIT RATINGS AFFILIATES ARE THEIR CURRENT OPINIONS OF THE RELATIVE FUTURE CREDIT RISK OF ENTITIES, CREDIT COMMITMENTS, OR DEBT OR DEBT-LIKE SECURITIES, AND MATERIALS, PRODUCTS, SERVICES AND INFORMATION PUBLISHED OR OTHERWISE MADE AVAILABLE BY MOODY'S (COLLECTIVELY, "MATERIALS") MAY INCLUDE SUCH CURRENT OPINIONS. MOODY'S DEFINES CREDIT RISK AS THE RISK THAT AN ENTITY MAY NOT MEET ITS CONTRACTUAL FINANCIAL OBLIGATIONS AS THEY COME DUE AND ANY ESTIMATED FINANCIAL LOSS IN THE EVENT OF DEFAULT OR IMPAIRMENT. SEE APPLICABLE MOODY'S RATING SYMBOLS AND DEFINITIONS PUBLICATION FOR INFORMATION ON THE TYPES OF CONTRACTUAL FINANCIAL OBLIGATIONS ADDRESSED BY MOODY'S CREDIT RATINGS. CREDIT RATINGS DO NOT ADDRESS ANY OTHER RISK, INCLUDING BUT NOT LIMITED TO: LIQUIDITY RISK, MARKET VALUE RISK, OR PRICE VOLATILITY. CREDIT RATINGS, NON-CREDIT ASSESSMENTS ("ASSESSMENTS"), AND OTHER OPINIONS INCLUDED IN MOODY'S MATERIALS ARE NOT STATEMENTS OF CURRENT OR HISTORICAL FACT. MOODY'S MATERIALS MAY ALSO INCLUDE QUANTITATIVE MODEL-BASED ESTIMATES OF CREDIT RISK AND RELATED OPINIONS OR COMMENTARY PUBLISHED BY MOODY'S ANALYTICS, INC. AND/OR ITS AFFILIATES. MOODY'S CREDIT RATINGS, ASSESSMENTS, OTHER OPINIONS AND MATERIALS DO NOT CONSTITUTE OR PROVIDE INVESTMENT OR FINANCIAL ADVICE, AND MOODY'S CREDIT RATINGS, ASSESSMENTS, OTHER OPINIONS AND MATERIALS ARE NOT AND DO NOT PROVIDE RECOMMENDATIONS TO PURCHASE, SELL, OR HOLD PARTICULAR SECURITIES. MOODY'S CREDIT RATINGS, ASSESSMENTS, OTHER OPINIONS AND MATERIALS DO NOT COMMENT ON THE SUITABILITY OF AN INVESTMENT FOR ANY PARTICULAR INVESTOR. MOODY'S ISSUES ITS CREDIT RATINGS, ASSESSMENTS AND OTHER OPINIONS AND PUBLISHES OR OTHERWISE MAKES AVAILABLE ITS MATERIALS WITH THE EXPECTATION AND UNDERSTANDING THAT EACH INVESTOR WILL, WITH DUE CARE, MAKE ITS OWN STUDY AND EVALUATION OF EACH SECURITY THAT IS UNDER CONSIDERATION FOR PURCHASE, HOLDING, OR SALE.
MOODY'S CREDIT RATINGS, ASSESSMENTS, OTHER OPINIONS, AND MATERIALS ARE NOT INTENDED FOR USE BY RETAIL INVESTORS AND IT WOULD BE RECKLESS AND INAPPROPRIATE FOR RETAIL INVESTORS TO USE MOODY'S CREDIT RATINGS, ASSESSMENTS, OTHER OPINIONS OR MATERIALS WHEN MAKING AN INVESTMENT DECISION. IF IN DOUBT YOU SHOULD CONTACT YOUR FINANCIAL OR OTHER PROFESSIONAL ADVISER.
ALL INFORMATION CONTAINED HEREIN IS PROTECTED BY LAW, INCLUDING BUT NOT LIMITED TO, COPYRIGHT LAW, AND NONE OF SUCH INFORMATION MAY BE COPIED OR OTHERWISE REPRODUCED, REPACKAGED, FURTHER TRANSMITTED, TRANSFERRED, DISSEMINATED, REDISTRIBUTED OR RESOLD, OR STORED FOR SUBSEQUENT USE FOR ANY SUCH PURPOSE, IN WHOLE OR IN PART, IN ANY FORM OR MANNER OR BY ANY MEANS WHATSOEVER, BY ANY PERSON WITHOUT MOODY'S PRIOR WRITTEN CONSENT. FOR CLARITY, NO INFORMATION CONTAINED HEREIN MAY BE USED TO DEVELOP, IMPROVE, TRAIN OR RETRAIN ANY SOFTWARE PROGRAM OR DATABASE, INCLUDING, BUT NOT LIMITED TO, FOR ANY ARTIFICIAL INTELLIGENCE, MACHINE LEARNING OR NATURAL LANGUAGE PROCESSING SOFTWARE, ALGORITHM, METHODOLOGY AND/OR MODEL.
MOODY'S CREDIT RATINGS, ASSESSMENTS, OTHER OPINIONS AND MATERIALS ARE NOT INTENDED FOR USE BY ANY PERSON AS A BENCHMARK AS THAT TERM IS DEFINED FOR REGULATORY PURPOSES AND MUST NOT BE USED IN ANY WAY THAT COULD RESULT IN THEM BEING CONSIDERED A BENCHMARK.
All information contained herein is obtained by MOODY'S from sources believed by it to be accurate and reliable. Because of the possibility of human or mechanical error as well as other factors, however, all information contained herein is provided "AS IS" without warranty of any kind. MOODY'S adopts all necessary measures so that the information it uses in assigning a credit rating is of sufficient quality and from sources MOODY'S considers to be reliable including, when appropriate, independent third-party sources. However, MOODY'S is not an auditor and cannot in every instance independently verify or validate information received in the credit rating process or in preparing its Materials.
To the extent permitted by law, MOODY'S and its directors, officers, employees, agents, representatives, licensors and suppliers disclaim liability to any person or entity for any indirect, special, consequential, or incidental losses or damages whatsoever arising from or in connection with the information contained herein or the use of or inability to use any such information, even if MOODY'S or any of its directors, officers, employees, agents, representatives, licensors or suppliers is advised in advance of the possibility of such losses or damages, including but not limited to: (a) any loss of present or prospective profits or (b) any loss or damage arising where the relevant financial instrument is not the subject of a particular credit rating assigned by MOODY'S.
To the extent permitted by law, MOODY'S and its directors, officers, employees, agents, representatives, licensors and suppliers disclaim liability for any direct or compensatory losses or damages caused to any person or entity, including but not limited to by any negligence (but excluding fraud, willful misconduct or any other type of liability that, for the avoidance of doubt, by law cannot be excluded) on the part of, or any contingency within or beyond the control of, MOODY'S or any of its directors, officers, employees, agents, representatives, licensors or suppliers, arising from or in connection with the information contained herein or the use of or inability to use any such information.
NO WARRANTY, EXPRESS OR IMPLIED, AS TO THE ACCURACY, TIMELINESS, COMPLETENESS, MERCHANTABILITY OR FITNESS FOR ANY PARTICULAR PURPOSE OF ANY CREDIT RATING, ASSESSMENT, OTHER OPINION OR INFORMATION IS GIVEN OR MADE BY MOODY'S IN ANY FORM OR MANNER WHATSOEVER.
Moody's Investors Service, Inc., a wholly-owned credit rating agency subsidiary of Moody's Corporation ("MCO"), hereby discloses that most issuers of debt securities (including corporate and municipal bonds, debentures, notes and commercial paper) and preferred stock rated by Moody's Investors Service, Inc. have, prior to assignment of any credit rating, agreed to pay Moody's Investors Service, Inc. for credit ratings opinions and services rendered by it. MCO and all MCO entities that issue ratings under the "Moody's Ratings" brand name ("Moody's Ratings"), also maintain policies and procedures to address the independence of Moody's Ratings' credit ratings and credit rating processes. Information regarding certain affiliations that may exist between directors of MCO and rated entities, and between entities who hold credit ratings from Moody's Investors Service, Inc. and have also publicly reported to the SEC an ownership interest in MCO of more than 5%, is posted annually at ir.moodys.com under the heading "Investor Relations — Corporate Governance — Charter and Governance Documents - Director and Shareholder Affiliation Policy."
Moody's SF Japan K.K., Moody's Local AR Agente de Calificación de Riesgo S.A., Moody's Local BR Agência de Classificação de Risco LTDA, Moody's Local MX S.A. de C.V, I.C.V., Moody's Local PE Clasificadora de Riesgo S.A., and Moody's Local PA Calificadora de Riesgo S.A. (collectively, the "Moody's Non-NRSRO CRAs") are all indirectly wholly-owned credit rating agency subsidiaries of MCO. None of the Moody's Non-NRSRO CRAs is a Nationally Recognized Statistical Rating Organization.
Additional terms for Australia only: Any publication into Australia of this document is pursuant to the Australian Financial Services License of MOODY'S affiliate, Moody's Investors Service Pty Limited ABN 61 003 399 657AFSL 336969 and/or Moody's Analytics Australia Pty Ltd ABN 94 105 136 972 AFSL 383569 (as applicable). This document is intended to be provided only to "wholesale clients" within the meaning of section 761G of the Corporations Act 2001. By continuing to access this document from within Australia, you represent to MOODY'S that you are, or are accessing the document as a representative of, a "wholesale client" and that neither you nor the entity you represent will directly or indirectly disseminate this document or its contents to "retail clients" within the meaning of section 761G of the Corporations Act 2001. MOODY'S credit rating is an opinion as to the creditworthiness of a debt obligation of the issuer, not on the equity securities of the issuer or any form of security that is available to retail investors.
Additional terms for India only: Moody's credit ratings, Assessments, other opinions and Materials are not intended to be and shall not be relied upon or used by any users located in India in relation to securities listed or proposed to be listed on Indian stock exchanges.
Additional terms with respect to Second Party Opinions and Net Zero Assessments (as defined in Moody's Ratings Rating Symbols and Definitions): Please note that neither a Second Party Opinion ("SPO") nor a Net Zero Assessment ("NZA") is a "credit rating". The issuance of SPOs and NZAs is not a regulated activity in many jurisdictions, including Singapore. JAPAN: In Japan, development and provision of SPOs and NZAs fall under the category of "Ancillary Businesses", not "Credit Rating Business", and are not subject to the regulations applicable to "Credit Rating Business" under the Financial Instruments and Exchange Act of Japan and its relevant regulation. PRC: Any SPO: (1) does not constitute a PRC Green Bond Assessment as defined under any relevant PRC laws or regulations; (2) cannot be included in any registration statement, offering circular, prospectus or any other documents submitted to the PRC regulatory authorities or otherwise used to satisfy any PRC regulatory disclosure requirement; and (3) cannot be used within the PRC for any regulatory purpose or for any other purpose which is not permitted under relevant PRC laws or regulations. For the purposes of this disclaimer, "PRC" refers to the mainland of the People's Republic of China, excluding Hong Kong, Macau and Taiwan.

OUTDATED METHODOLOGY

This publication does not announce a credit rating action. For any credit ratings referenced in this publication, please see the issuer/deal page on https://ratings.moodys.com for the most updated credit rating action information and rating history.

REPORT NUMBER 1400901

OUTDATED METHODOLOGY

Authors

Dirk Steinicke
Ryohei Nishio, CFA
Anastasija Johnson

OUTDATED METHODOLOGY